



AWS Lake Formation

Lecture Slides

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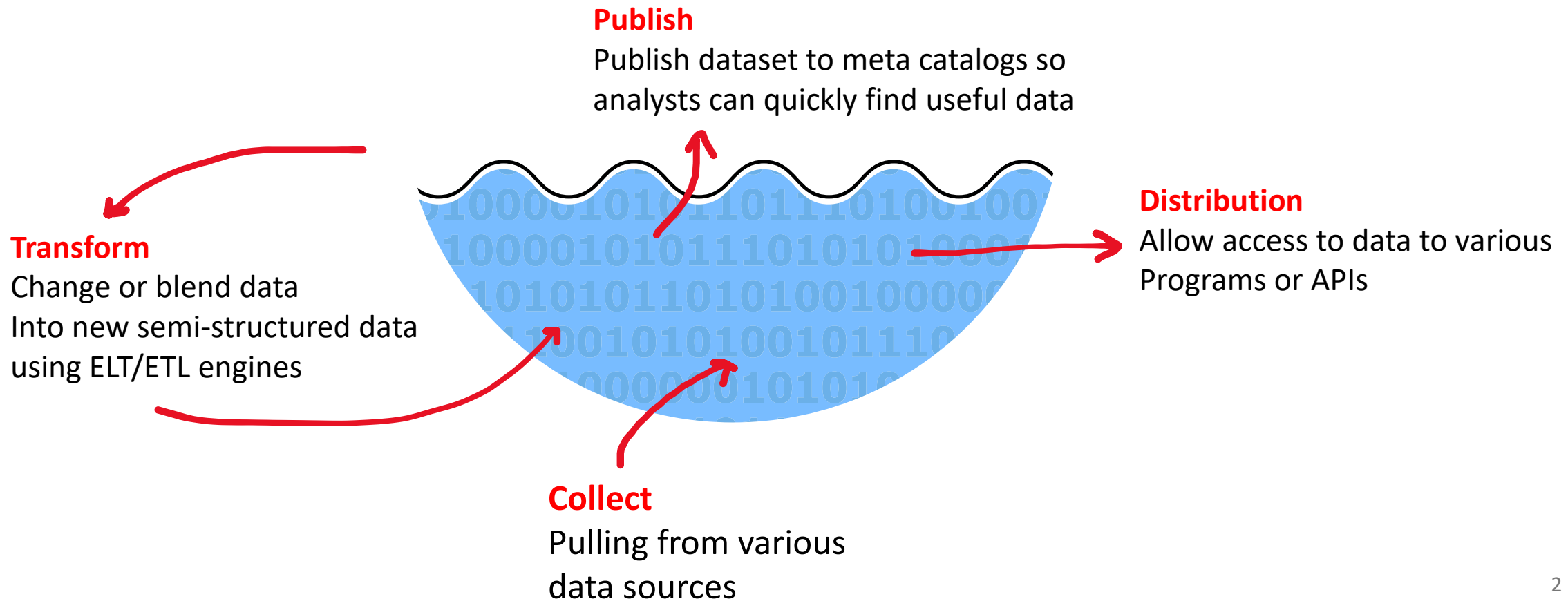
Good luck on your exam!

Introduction to Data Lakes

A data lake is a **centralized data repository for unstructured and semi-structured data**

A Data Lake is intended to store vast amounts of data

Data lakes generally use **object (blobs) or files** as its storage medium.



AWS Lake Formation



AWS Lake Formation is **a data lake** to centrally govern, secure, and globally share data for analytics and machine learning

- Manage fine-grained access control for your data lake data on Amazon S3
- Manage metadata in AWS Glue Data Catalog
- Lake Formation provides its own permissions model that augments the IAM permissions model
 - through a simple grant or revoke mechanism similar to relational database management system (RDBMS)
- allows you to share data internally and externally across multiple AWS accounts, AWS organizations or directly with IAM principals in another account
- permissions are enforced using granular controls at the column, row, and cell-levels across
 - Athena
 - Quicksight
 - Redshift Spectrum
 - EMR
 - Glue



AWS Lake Formation and AWS Glue share the same Data Catalog

AWS Lake Formation

Data Lake is your persistent data that is stored in Amazon S3 and managed by Lake Formation using a Data Catalog.

- Structured and unstructured data
- Raw data and transformed data

Blueprint a data management template that enables you to easily ingest data into a data lake. From a blueprint, you can create a workflow.

Workflow is a container for a set of related AWS Glue jobs, crawlers, and triggers.

Data Catalog is your persistent metadata store.
It is a managed service that lets you store, annotate, and share metadata

Data lake administrator is a principal who can grant any principal (including self) any permission on any Data Catalog resource or data location.